

Module 2

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Module 2 Study Guide and Deliverables

Readings:	• Review Houser, Chapter 12 (pages 454-498) • Review Promoting Interoperability component of MIP • Review Overview of Quality Payment Program (QPP) • Recent article – will be posted in the Live Classroom Slides and Materials discussion forum
Assignments:	Assignment 2 due Tuesday, January 28 at 6:00 AM ET
Assessments:	Quiz 2 due Tuesday, January 28 at 6:00 AM ET
Discussions:	For grading consideration, all posts to this discussion topic must be made prior to Tuesday, January 28 at 6:00 AM ET. Any posts after this date/time will not be included in the grading process.
Live Classroom:	• Wednesday, January 22 at 8:00 PM ET • Saturday, January 25 at 12:00 PM (noon) ET • Facilitator Session: Thursday, January 23 at 8:30 PM ET

Module Learning Objectives

After successfully completing this module, participants will have learned:

- What are the components of an HIS/EHR system
- Ambulatory HIS/EHR functionality
- Acute care HIS/EHR and Hospital Information System functionality
- How an EHR works through hands-on use of an EHR system
- What are the federal government regulations related to EHRs
- Federal government certification requirements for an EHR system

Lecture 4: HIS/EHR Functional Overview and Clinical Functionality

Applying an Enterprise Architecture to an HIS/EHR

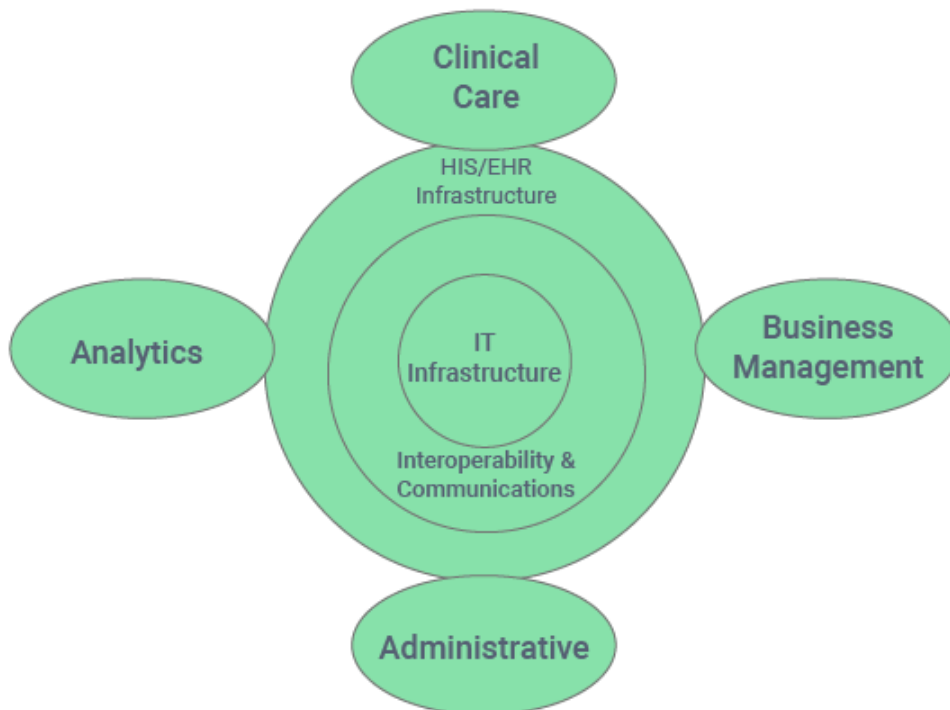
An EA can be valuable to an HIS/EHR implementation as it provides a methodology to incorporate all elements of the healthcare workflow into the requirements and then the technical infrastructure design. This approach is then integrated with the performance requirements for the HIS/EHR, which will be covered in a later lecture. The result is a clear understanding of the requirements for the technical infrastructure.

In applying an EA methodology to an HIS/EHR implementation, you strive to answer these questions:

1. What are the subcomponents for an HIS/EHR Enterprise Architecture?
2. What data might you need to determine the Enterprise Architecture?
3. How would you get this data?
4. Who supplies the data?

Going back to the Balanced Performance Model (BPM), let's focus on the "technology" component—specifically, the functionality of an HIS/EHR system. A HIS/EHR system is a database software system that manages multiple aspects of healthcare activities and healthcare workflow, supported by a sophisticated IT infrastructure as follows.

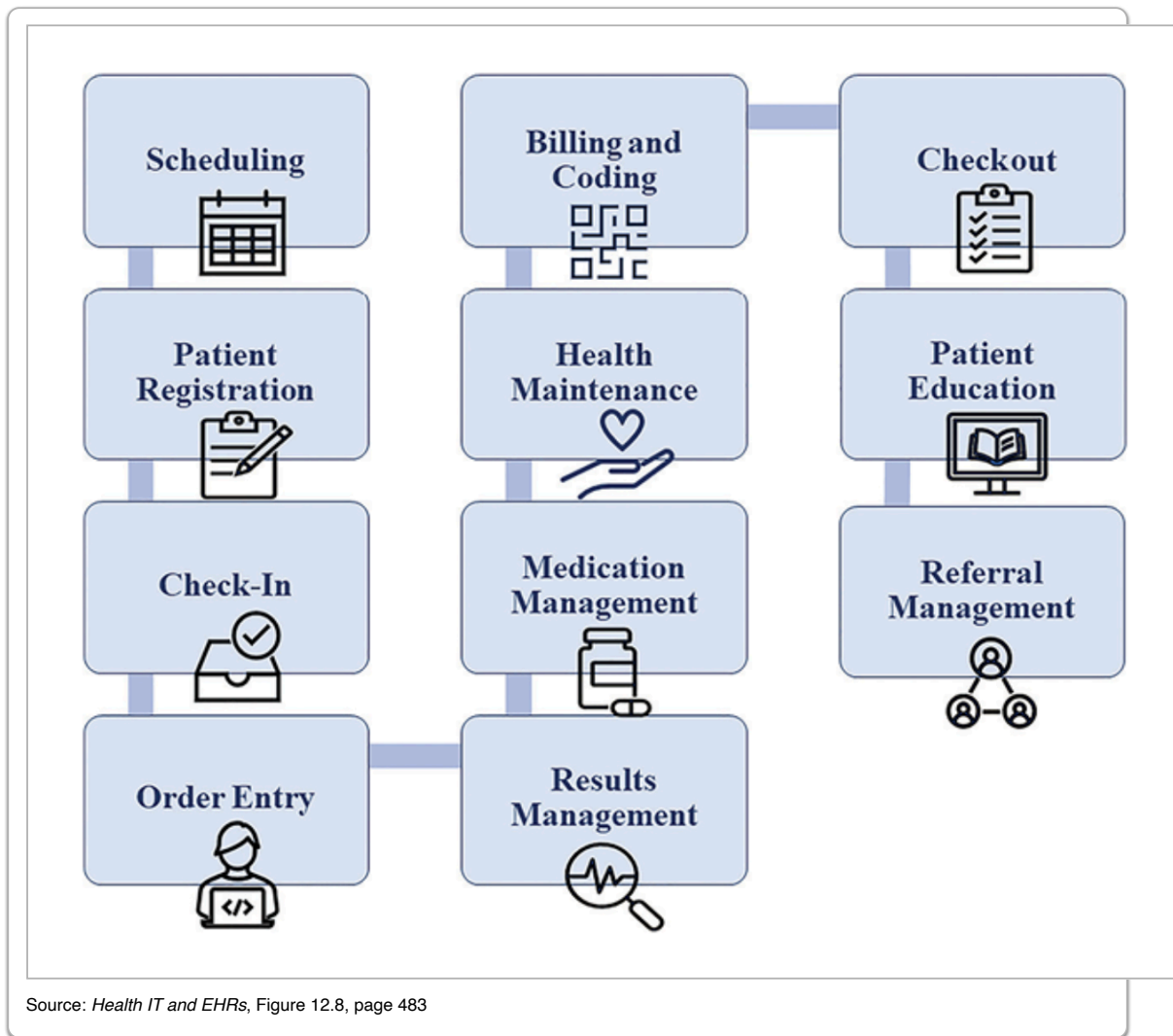
HIS/EHR Functionality



The textbook describes this complex, integrated environment and system in Chapter 1 and illustrates it in Figures 1.6 and 1.7. To understand HIS/EHR functionality we will use an EA framework. This module covers the business requirements, data needed, and application functionality. We will cover the underlying technical infrastructure in a separate module.

HIS/EHR Business Requirements

As described previously, HIS/EHR systems are used in multiple healthcare provider environments. Increasingly the systems are integrated to better track and coordinate care across them. The Business Layer requirements of a HIS/EHR vary depending on the type of provider setting. For example, clinical functionality in ambulatory care includes the key processes shown in Figure 12.8 of the textbook:.



These clinical functions are further detail into

- Clinical documentation
- Order entry and management
- Decision support tools
- Patient management
- Billing and coding
- Patient engagement
- Quality improvement
- Security and compliance

See the textbook pages 479-480 for more about these functions.

Similarly, the textbook provides detail on acute care requirements in section 12.2, pages 454-478 and specialty care settings on pages 492-498.

There is of course much more detail about these different healthcare business activities and processes. As we did in the lecture on workflow, we will discuss a sample of them in-depth. Review and discussion of all of them is out of the scope of this course.

HIS/EHR Data

Given the complex business requirements, the data incorporated into a HIS/EHR is complex. It reflects the data needed and created by the business activities. As one example, let's review the data needed for clinical care. EHR systems are built to aggregate and share information with other healthcare providers and organizations—such as laboratories, specialists, medical imaging facilities, pharmacies, emergency facilities, and school and workplace clinics—so they contain information from all clinicians involved in a patient's care. That data includes:

- Administrative and billing data
- Patient demographics
- Progress notes
- Vital signs
- Medical histories
- Diagnoses
- Medications
- Immunization dates
- Allergies
- Radiology images
- Lab and test results

Source: <https://www.healthit.gov/faq/what-information-does-electronic-health-record-ehr-contain>

Successful clinical care incorporates other data as well. Some examples are:

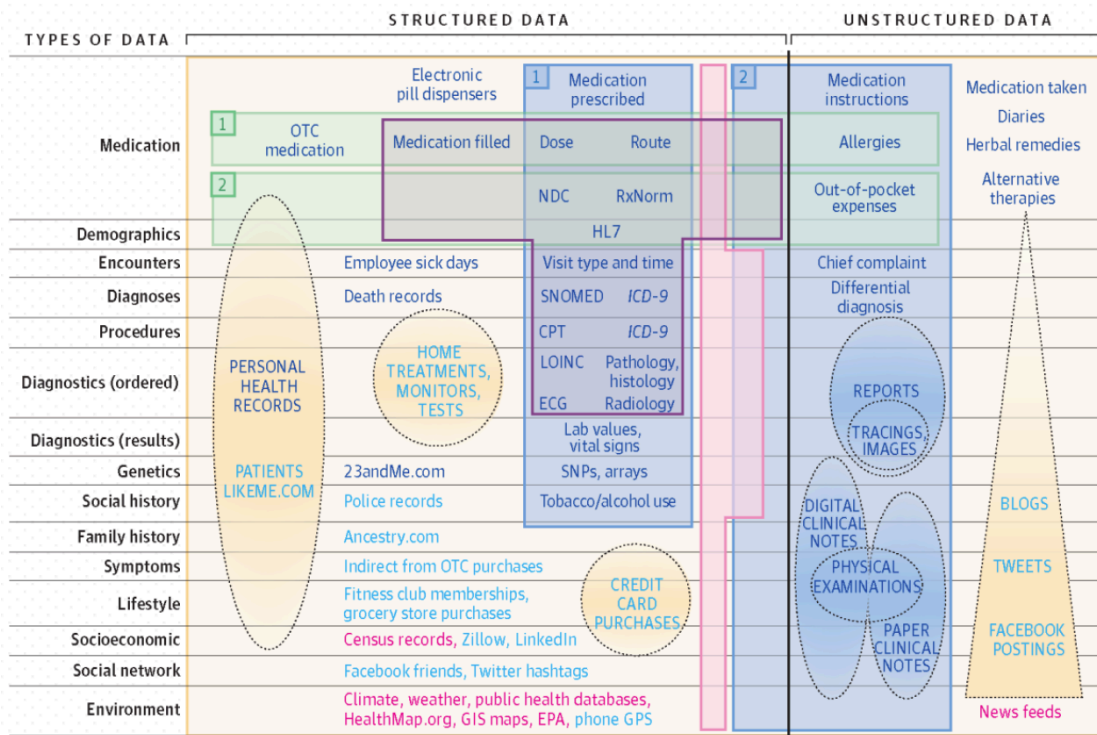
- Claims Data—includes patient demographics, diagnosis codes, dates of service, and the cost of services
- Social and Community Determinants of Health—includes community and social traits, such as average incomes, English proficiency, local healthy food choices, violence rates, transportation access, unemployment rates, and education
- Patient-Generated Health Data (PGHD)—PGHD takes many forms, from satisfaction surveys and patient-reported outcomes (PROs) to communications through a patient portal and data streaming from wearable fitness trackers and other Internet of Things (IoT) devices

Source: <https://www.techtarget.com/healthtechnanalytics/news/366591849/Which-Healthcare-Data-is-Important-for-Population-Health-Management>

Other HIS/EHR functions have equally diverse data. For example, patient scheduling requires knowing patient demographics, provider information, schedules, location, reasons for the encounter, etc.

HIS/EHR data is both structured and unstructured. Structured information includes a list of patient medications or details about patient billing charges. Unstructured data includes images and video, such as radiology; text, such as transcription reports; or sounds, such as recordings of fetal heartbeats.

One diagram that shows this complex data environment is:

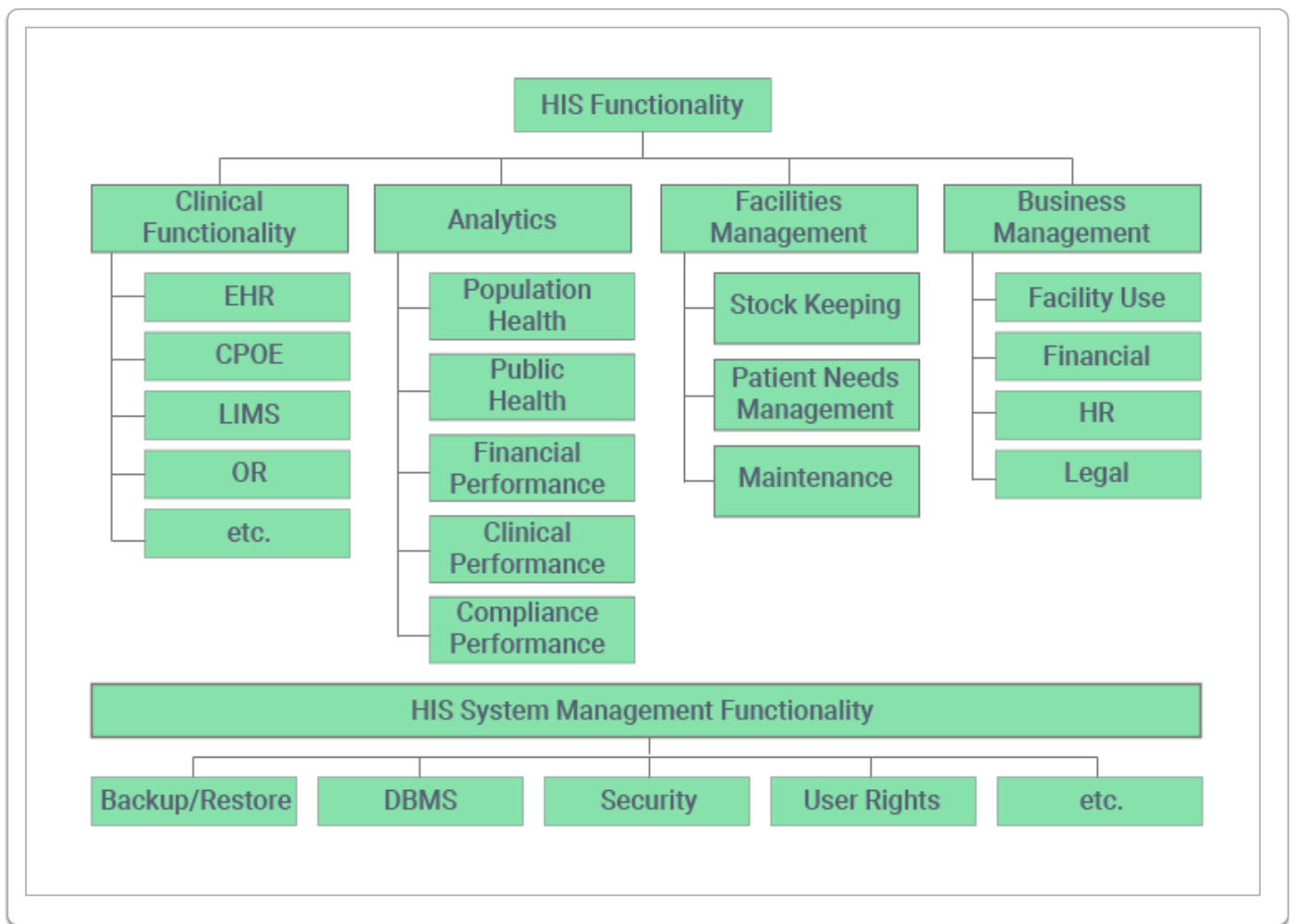


HIS/EHR Application Functionality

An HIS/EHR has functionality in four broad areas:

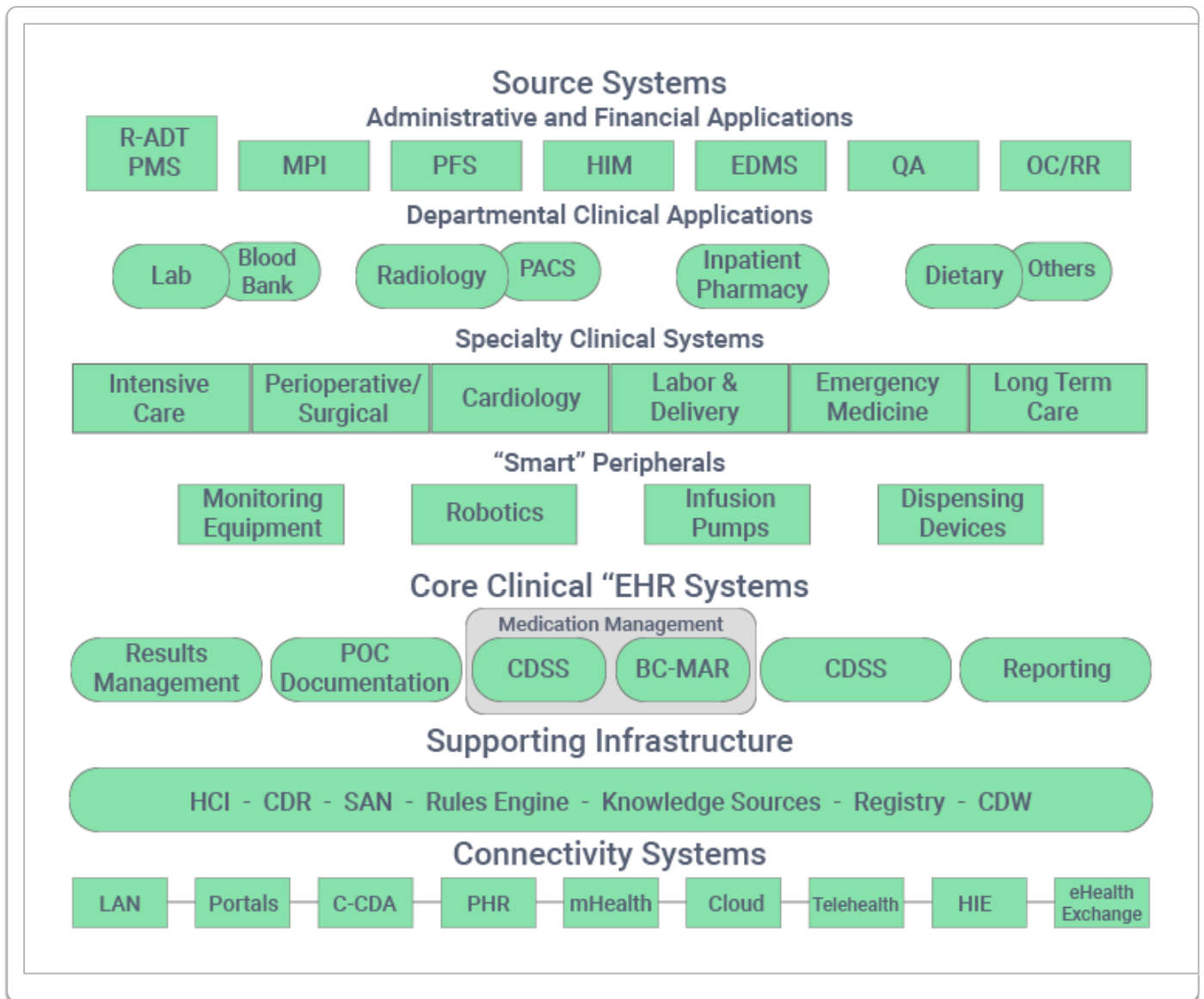
- Clinical Care—tracking and managing clinical activities
- Analytics—understanding the health and the effectiveness and cost of treatment of groups of patients
- Business Management—managing the finances and especially the revenue of a healthcare provider
- Administrative Management—managing the non-clinical activities of a healthcare provider

The following diagram shows key components of the HIS/EHR Application Functionality—the application layer of an EA.



In this lecture, we focus on clinical functionality. In a later lecture, we will review aspects of analytic and business management functionality. All are supported by IT infrastructure, which we will also cover in later lectures.

Given the many business requirements an HIS/EHR system might need to address, the systems are often complex as shown in the textbook, Chapter 1, Figure 1.6.



At a high level, a clinical EHR system enables use of data at the point of care and is the core system supporting clinical processes. It uses information from source systems and is integrated with departmental clinical applications and specialty clinical systems that provide area-specific functionality. Increasingly, these systems are connected to "smart" peripherals and mHealth devices. Finally, an HIS/EHR system requires the underlying capabilities of the supporting infrastructure and connectivity or interoperability systems. This and other MET courses discuss the different system elements as follows.

- Clinical Care (Core Clinical EHR Systems): Covered in this lecture
- Departmental Clinical Applications and Specialty Clinical Systems: We do not cover this in detail in this course, but it is partly covered in the textbook Chapter 12
- Source Systems: We do not cover this in detail in this course, but it is partly covered in the textbook
- Supporting Infrastructure: Covered in Lectures 9 and 10
- Connectivity Systems or Interoperability: Covered in Lectures 11 and 12 and in MET CS781
- "Smart" Systems: Covered in MET CS781

EHR and HIS systems are often segregated by the type of healthcare provider organization using the system. Major organizational types include:

- Acute: Hospital settings—often called *in-patient* or *acute care providers*

- Post-acute: Still an in-patient provider but less intense than a hospital, such as a rehabilitation hospital
- Long-term: Typically, residential but designed for long-term residence, such as a nursing home
- Ambulatory: Non-residential clinical settings such as doctors' offices and clinics
- Home: Care provided in the home

This course focuses on ambulatory and acute. The others are partly discussed in Chapter 12 in the textbook.

Health Level Seven International (HL7)

"Founded in 1987, Health Level Seven International (HL7) is a not-for-profit, ANSI-accredited standards developing organization. HL7 is dedicated to providing a comprehensive framework and related standards for the exchange, integration, sharing, and retrieval of electronic health information that supports clinical practice and the management, delivery, and evaluation of health services. HL7 has more than 1,600 members in 50 countries, including approximately 500 corporate members representing healthcare providers, government stakeholders, payers, pharmaceutical companies, vendors/suppliers, and consulting firms."

(Source: [About Health Level 7 International](#))

HL7 provides multiple models and standards related to EHRs and interoperability including:

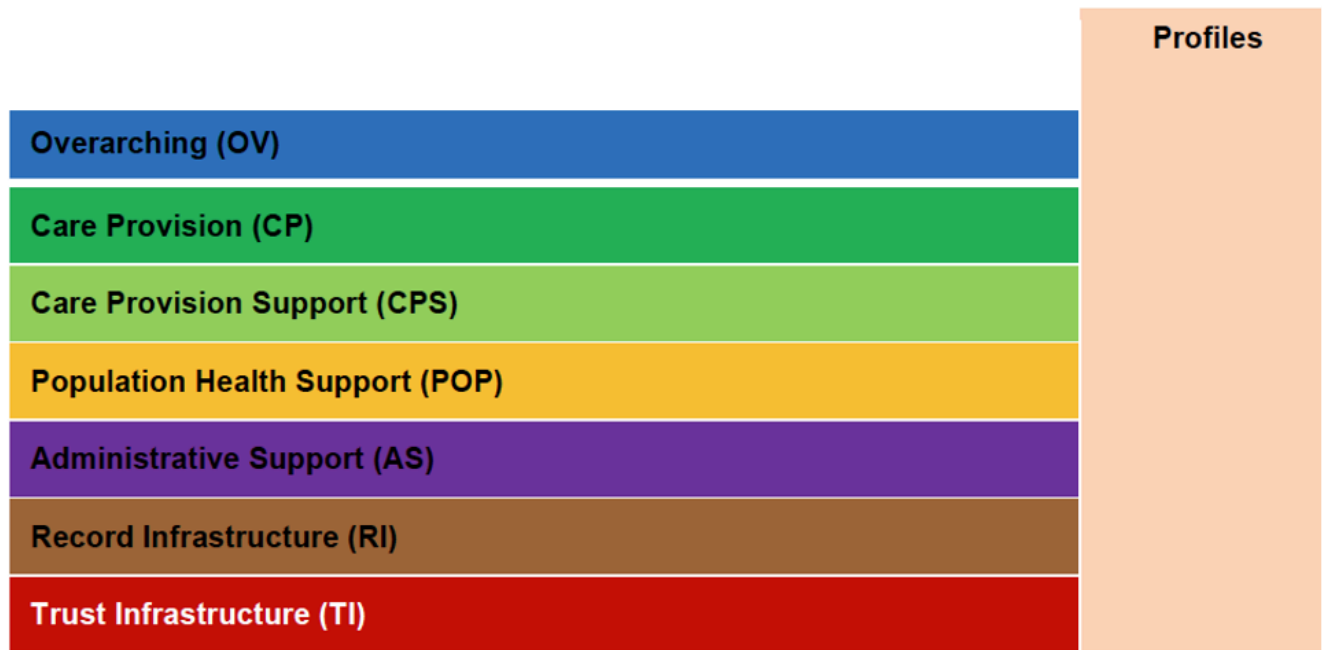
- EHR functional profiles and models that define EHR functionality both generally and related to specific functional areas and specific medical practices (such as child health)
- Interoperability protocols and standards such as a Clinical Document Architecture (CDA) and Fast Healthcare Interoperability Resources (FHIR) to facilitate the sharing of clinical data
- Implementation guides to assist in the application of HL7 standards

HL7 complements these models with education, documentation, and other materials to support use of HL7 standards.

The core HL7 description of EHR functions is called *HL7 EHR-System Functional Model, Release 2*. This model provides a high-level structure for EHR functionality. These are further broken down into subcomponents which are described in Release 3.

A diagram of the HL7 EHR Functional Model Release 2 follows. It is comprised of the seven sections shown. These are in turn applied to different "profiles" often representing different types of healthcare providers.

HL7 EHR Functional Model Release 2



HL7 EHR System Functional Model Release 2, Copyright © 2014 Health Level Seven International

The functionality described in Release 2 is further broken down by types of healthcare providers or activities. The types include:

- Behavioral Health
- Child Health
- Clinical Research
- Pharmacist/Pharmacy
- Records Management and Evidentiary Support
- System Long-Term Care
- ePrescribing
- Public Health
- Personal Health Record System Functional Model

In this course, we will explore two functional areas in more detail: ambulatory EHR system functionality and acute system functionality.

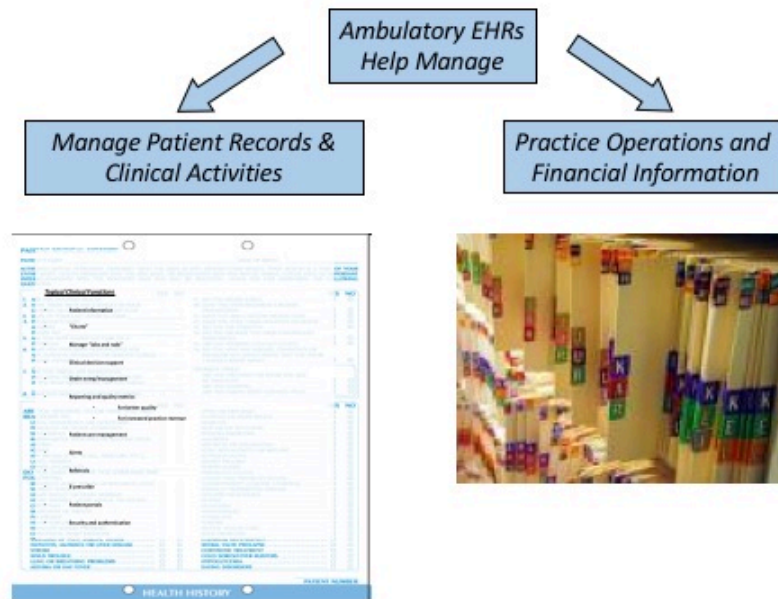
This description is just the surface of HL7's model for EHR functionality. The official documents that describe its standards are detailed and extensive.

See more at:

- [Health Level 7 International](#)
- [EHR - Electronic Health Records](#)

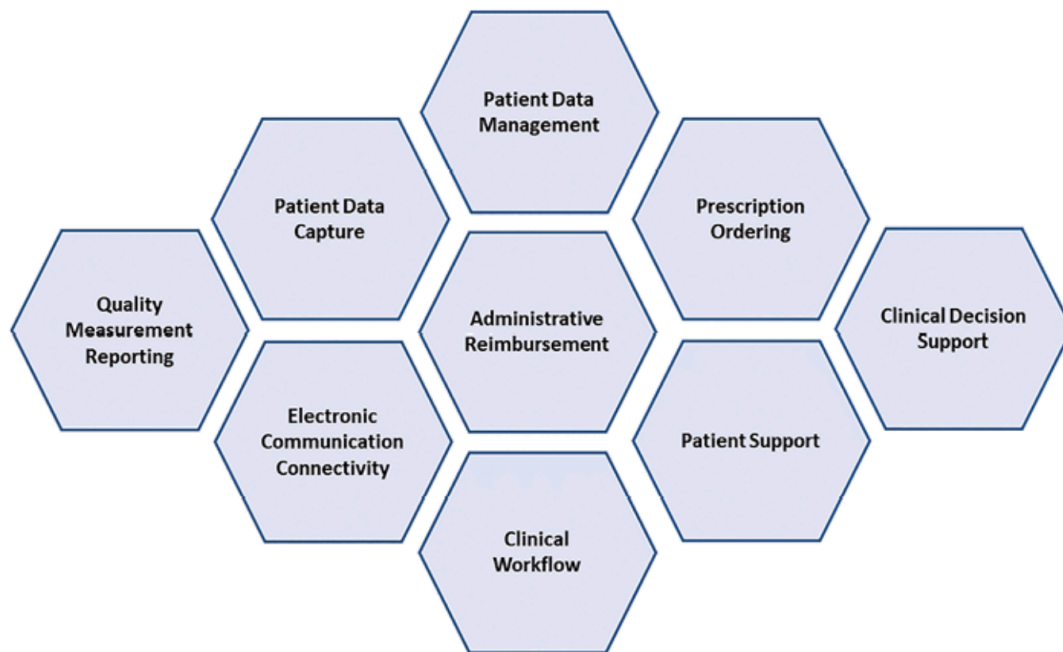
Ambulatory EHR Functionality

Ambulatory care is typified by numerous patient encounters often over a long period of time. To be useful in this setting, an EHR must support reviewing past information to see patterns that might suggest a diagnosis or to help manage chronic conditions. Ambulatory EHRs provide both clinical and business functionality. The former relates to managing patient clinical data while supporting the business of an ambulatory practice.



Ambulatory EHR systems often also need other functionality for a variety of specialties. For example, ophthalmology, orthopedics, and psychology all require very different information. Ambulatory EHR systems require interoperability and security functionality as defined in the HL7 model.

Again, key ambulatory processes are shown in Figure 12.8 of the textbook. They lead to EHR functional requirements as shown in Figure 12.9.



Source: *Health IT and EHRs*, Figure 12.9, page 484

Both are detailed in the textbook *Health IT and EHRs*, Chapter 12.

An ambulatory EHR must also have the security features that include:

- Authentication of users (proving identity)
- Controlling access based on user role or the context of a care situation
- Auditing access and use of records
- Encryption of any data sent out of the system
- Protection against viruses and other malware
- Backup of data to prevent loss in case of computer failure or disaster

HIPAA and other security regulations create need for extensive security functionality. Given the importance of security, this topic is covered in detail in a separate lecture.

As the need for interoperability between healthcare providers and organizations increases, HL7 has been a leader in defining structures to share information which we will review in a later lecture.

Acute EHR Functionality

Acute EHRs incorporate much of the functionality of ambulatory EHRs, especially as it relates to tracking patient encounter results (maintaining an electronic "chart"). But acute systems must also support:

- Even more complex interoperability requirements, both within the hospital or acute setting and across multiple care provider organizations,
- More types of departmental and specialty requirements, and

- More complex non-clinical activities, such as bed and stock management.

Therefore, acute EHRs can be thought of as being a part of a larger, more complex hospital information system (HIS). These functions include:

- Results management
- Laboratory test results
- Diagnostic imaging reports
- Point of care documentation
 - Nursing documentation
 - Physician documentation
- Medication management
- Computerized Physician Order Entry (CPOE)
- Clinical decision support
- “Bed” management (think of a hotel management system)
- Inventory and stock management (think of a supply chain management system)
- Many other functions

For details on many of these functions, see the lectures on non-clinical functions and the Chapter 12 (pages 454-471) of the textbook, *Health IT and EHRs*.

Clinical Decision Support Systems

An important clinical function is supporting clinical decisions, also called *Clinical Decision Support (CDS)*. The Agency for Healthcare Research and Quality (AHRQ) defines CDS as systems that “provide clinicians, staff, patients, and other individuals with knowledge and person-specific information, intelligently filtered and presented at appropriate times, to enhance health and healthcare.” This information supports physicians in proper diagnosis and treatment of health problems.”

Source: Clinical Decision Support Systems: State of the Art

AHRQ Publication No. 09-0069-EF

June 2009

CDS systems typically provide the following functions:

- Alerting
- Reminding
- Critiquing
- Interpreting
- Predicting
- Diagnosing
- Assisting
- Suggesting

Type of CDS Tool	Description
Alerts and reminders	Notify providers about critical info such as drug interactions, allergies, overdue tests, or guideline deviations.
Clinical guidelines	Provide evidence-based protocols to guide diagnosis and management of conditions.
Diagnostic support tools	Analyze patient data to suggest possible diagnoses.
Condition-specific order sets	Standardized sets of orders for specific conditions to ensure consistent care.
Documentation templates	Standardize the documentation process for accuracy and completeness.
Patient data reports and summaries	Aggregate and display patient data in a structured manner for review and decision-making.
Dose calculation support	Calculate drug dosages based on patient-specific information, such as weight and age.
Clinical pathways	Provide step-by-step care guidance for specific conditions, integrating best practices.
Drug formulary guidance	Offer information on drug choices, formulary status, and insurance coverage to support prescribing decisions.
Patient-specific alerts and recommendations	Generate tailored alerts and recommendations based on the patient's individual context.
Advanced analytics and predictive modeling	Use algorithms and machine learning to analyze data for trends, predict outcomes, and suggest preventative measures.

Source: *Health IT and EHRs*, Table 12.6, page 472

CDS systems can be valuable elements of an EHR, providing clinicians important help to effectively treat patients, including increasing the accuracy of diagnoses, identifying optimal treatment plans, and reducing the possibility of errors. At the same time, CDS systems can hinder effective care practices or even lead to increased errors. It is critical that CDS systems add value to clinical processes. Whether CDS systems are a bane or a benefit depends on successful blending of four requirements:

1. **Accurate**—Clinical decision support starts with clinical knowledge—how to treat specific health problems in patients. This information should be the best available evidence on the topic at hand.
2. **Patient Related**—A CDS should provide information that is relevant and specific to the particular patient.
3. **Filtered**—A CDS should filter the information to avoid extraneous information. For example, when working in pediatrics, doctors don't need to have clinical knowledge about diseases exclusive to the elderly population.
4. **Contextual**—Information should enhance clinical efforts by being presented when needed and in a format that is easy to understand. It should respond to its context in the workflow.

These four work together so the CDS helps the clinician with the right information, presented in the right way, and at the right time in the clinical process.

There are multiple types of CDS systems ranging from simple to very complex. We can categorize them into the following major types.

KINDS	WHAT IT DOES	HOW IT WORKS
Data-driven	Makes decisions based on data from internal databases or external databases.	Uses data mining techniques to discern trends and patterns for predicting future events. Often used to help make decisions about inventory, sales and other business processes.
Model-driven	Customized according to a predefined set of user requirements.	Used to analyze different scenarios to meet user requirements—for example, assisting with scheduling or developing financial statements.
Communication-driven and group	Uses a variety of communication tools to allow more than one person to work on the same task.	Increases collaboration between users and the system; improves overall efficiency and effectiveness of a system.
Knowledge-driven	Data resides in a continuously updated knowledge base that's maintained by a knowledge management system.	Provides data to users that's consistent with a company's business processes and knowledge base.
Document-driven	Type of information management system that uses documents to retrieve data.	Enables users to search webpages or databases, or find specific search terms, such as those related to policies and procedures, meeting minutes and corporate records.

Source: <https://healthit.ahrq.gov/ahrq-funded-projects/current-health-it-priorities/clinical-decision-support-cds/chapter-1-approaching-clinical-decision/section-4-types-cds-interventions>

Clinical decision support is an area of HIS/EHRs that is especially being impacted by the expanding use of artificial intelligence in healthcare. The textbook provides more discussion of CDS on pages 29-30 and 471-475. It goes into detail about AI in Chapter 15. This topic is covered in more depth in MET CS 781.

Using an EHR

To have a real understanding of how an EHR system works, it's important to try it for yourself. In this course, we use a product called *dr chromo* — a free, web-based EHR system. *dr chromo* is especially focused on the needs of small to mid-size ambulatory physician practices; it does not have the full functionality of a HIS. But since almost all of us are at some time patients, we generally understand the activities of an ambulatory practice so *dr chromo* is a good learning tool. If you have access to some other EHR or to an HIS, by all means explore its functionality and please share it with the class.

To use *dr chromo*, go to <https://www.drchrono.com/university-ehr-emr/>. To create your account, click on "Student Sign Up " link and fill in the form. Leave the "Enter or select a specialty" field blank and say 1-2 providers on the next field. Provide a Username and Password per the next screen By submitting these forms you will be agreeing to the Privacy Policy and Terms of Use.

Follow the attached *dr chrono* Script as if you were a physician and go through the process of using an EHR. The *dr chrono* website can provide you with more information to deepen your understanding and experience.

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■ Lecture 5: Federal Government EHR Regulations and Certification

Federal Government EHR Regulations

The federal government has taken multiple actions to promote and incent the use of EHRs. Its goals are described as follows:

"The Centers for Medicare & Medicaid Services (CMS) encourage eligible clinicians, eligible hospitals, and CAHs to adopt and meaningfully use certified electronic health record (EHR) technology through the Promoting Interoperability Programs. In connection with other CMS initiatives such as the Quality Payment Program, these programs help to support high quality, high value care through the use of certified health IT." Meaningful Use is described by ONC [here](#). Certification is discussed in this lecture. The federal government does this by promoting Meaningful Use of an EHR. The goals of meaningful use are:

1. Improving quality, safety, efficiency, and reducing health disparities
2. Engage patients and families in their healthcare
3. Improve care coordination
4. Ensure privacy and security protection for personal health information
5. Improve population and public health

Source: <https://www.healthit.gov/topic/meaningful-use-and-macra/promoting-interoperability>

The federal efforts focus on healthcare delivery. They do not include goals related to the business of healthcare, even though EHR and HIS functionality are used for that purpose as well.

Under the current federal programs, the government encourages—if not mandates—the use of EHRs. Not meeting federal regulations related to the use of an EHR could result in a substantial reduction of Medicare and Medicaid reimbursements.

Federal Regulations of EHRs: MACRA, QPP, MIPS, PI and APMs

The initial federal program related to EHR use was a provision in the American Recovery and Reinvestment Act (ARRA) of 2009, called *The Health Information Technology for Economic and Clinical Health (HITECH) Act*. The act contained health information technology (HIT) including incentive payments earmarked for adoption and deployment of EHRs. It committed to spending \$49 billion, an unprecedented monetary commitment from the government to make EHR and healthcare IT adoption a priority in the United States.

The HITECH Act has ended. The federal government has implemented a replacement called the *Quality Payment Program (QPP)*.

Common Healthcare and Healthcare Information Technology Acronyms

- ACI—Advancing Care Information
- ACO—Accountable Care Organization—a healthcare delivery organization managing shared risk
- APM—Alternative Payment Model—a federal or private payer reimbursement model to implement shared risk
- ARRA—American Recovery and Reinvestment Act
- CAH—Critical Access Hospital
- CDS or CDSS—Clinical Decision Support System
- CPOE—Computerized Provider Order Entry
- CMS—Centers for Medicare and Medicaid Services
- CQM—Clinical Quality Measure
- ECP—Eligible Care Providers
- EH—Eligible Hospital
- eHealth—Electronic delivery of healthcare or electronic technology to support healthcare
- EHR—Electronic Health Records
- EMR—Electronic Medical Records
- EP—Eligible Professional
- ePrescribe or eRx—Generate and transmit permissible prescriptions electronically
- HHS—The U.S. Department of Health and Human Services
- HIE—Health Information Exchange
- HIS—Hospital Information System
- HIT—Healthcare Information Technology
- HITECH—Health Information Technology for Economic and Clinical Health Act
- LIMS—Laboratory Information Management System
- MACRA—Medicare Access and CHIP Reauthorization Act of 2015
- Managed Care—Another term for Shared Risk

- mHealth—Use of mobile devices to deliver or support healthcare
- MIPS—Merit-Based Incentive Payments System
- ONC—Office of the National Coordinator
- ONCHIT—Office of the National Coordinator of HIT
- PACS—Picture Archiving and Communication System—a HIS system to manage healthcare images like X-rays.
- PCMH—Patient-Centered Medical Home
- PHR—Personal Health Record
- PI—Promoting Interoperability
- Shared Risk—A new type of healthcare reimbursement approach where providers share in cost savings and overruns
- Value-Based Care—Another term for shared risk

The MACRA Quality Payment Program (QPP) continues changing Medicare to a new business model variously called *value-based*, *managed care*, *shared risk*, or *alternative payment models*, which links quality and value to payments and provides incentives for cost savings.

Under QPP, providers participate in one of two tracks:

1. Merit-Based Incentive Payment System (MIPS), or
2. Advanced Alternative Payment Models (APMs), which are innovative payment models emphasizing the federal efforts to change the overall healthcare business model to a new approach.

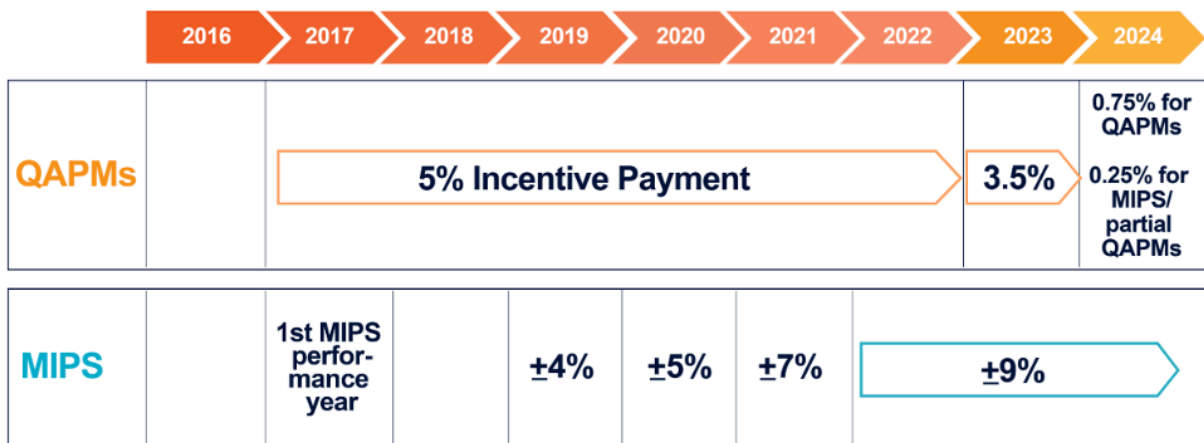
Two Participation Tracks



In addition, MIPS has several reporting frameworks.

Under MIPS, CMS scores clinicians and groups on their performance in four categories, with the following weights for 2024:

2024 MIPS Scoring Components and Weights



Source: <https://qpp.cms.gov/resources/link/dafeaf87-2237-4427-9107-508624eb315f>

Under Medicare, there are also multiple APM models. In general, they incent providers to deliver quality care cost effectively by “sharing risk.” Providers have a target amount of dollars to provide care for a group of patients. If the providers meet quality goals at a lower cost than the target, they receive some of the savings. If they miss the quality or cost goals, their Medicare reimbursement is lowered. In most cases, use of an EHR is part of the APM program’s goals. Given the complexity of APMs, this course focuses on MIPS.

Promoting Interoperability (PI) is the component of MIPS/QPP that relates to EHRs. In 2024, PI has a weight of 25% to 30% of a provider’s MIPS score giving it large impact. PI continues to require healthcare providers to use an EHR in a “meaningful” fashion. Providers are required to use an EHR for functions that both improve and demonstrate the quality of care, such as e-prescribing, electronic exchange of health information, and submission of quality measures. In addition, the EHR must be certified. Certification of an EHR system assures organizations that it offers the capability, functionality, and security to meet the federal criteria. More detail about certification is later in this lecture.

This PI score is calculated based on participants submitting multiple measures of EHR use and attesting to other measures of EHR use.

For 2024, providers must submit collected data for certain measures from each of five objectives.

- Electronic Clinical Quality Measures (eCQMs),
- MIPS Clinical Quality Measures (CQMs),
- Qualified Clinical Data Registry (QCDR) Measures,
- Medicare Part B Claims Measures, and
- [The Consumer Assessment of Healthcare Providers and Systems \(CAHPS\) for MIPS Survey.](#)

In addition to submitting measures, you must provide your EHR’s CMS identification code from the [Certified Health IT Product List \(CHPL\)](#) and submit a “yes” to:

- The Actions to Limit or Restrict Compatibility or Interoperability of CEHRT (previously named the Prevention of Information Blocking) Attestation,
- The ONC Direct Review Attestation,
- The security risk analysis measure, and
- The Safety Assurance Factors for EHR Resilience (SAFER) Guides Measure.

There are many other details about the MIPS/PI which can be found at [Promoting Interoperability Measures: Traditional MIPS Requirements](#), and for 2024, at <https://qpp.cms.gov/mips/promoting-interoperability?py=2024>.

Federal Regulations of EHRs: The 21st Century Cures Act

The 21st Century Cures Act, signed December 13, 2016, by President Obama, promotes and funds the acceleration of research into preventing and curing serious illnesses; accelerates drug and medical device development; attempts to address the opioid abuse crisis; and tries to improve mental health service delivery. The Act expands federal regulations especially related to greater interoperability, adoption of electronic health records (EHRs), and management of information blocking.

Certain sections are focused on "improving quality of care for patients," with interoperability a main concern. It also places strong emphasis on providing patients access to their electronic health information that is "easy to understand, secure, and updated automatically."

Source: <https://www.himss.org/resources/21st-century-cures-act-summary>

The federal government is doing much to implement the HIS/EHR components of the Cures Act. Details can be seen at:

- [Health IT Legislation](#)
- [21st Century Cures Act: Interoperability, Information Blocking, and the ONC Health IT Certification Program](#)

Federal EHR Program Audits

CMS performs pre-payment and post-payment audits to ensure that federal money has been spent appropriately. CMS therefore states that "Providers should retain ALL relevant supporting documentation (in either paper or electronic format) used in the completion of the Attestation Module responses for six years post-attestation."

CMS also allows for hardship exemptions to avoid penalties. You can see details on the [2024 MIPS Data Validation Criteria](#) page at CMS's website.

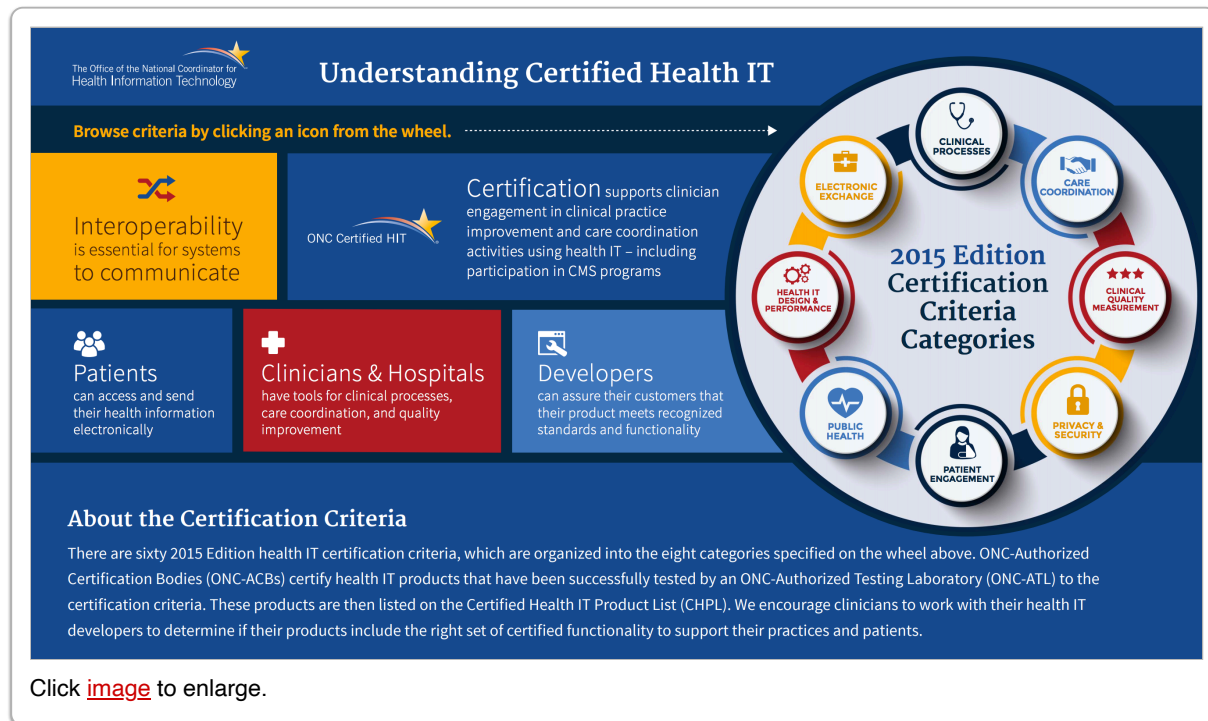
In addition, EHRs are subject to security and privacy audits to help ensure that these important requirements are being met. We will discuss these requirements in later lectures.

Federal EHR Certification

Another aspect of government regulation of EHRs is EHR Certification. Certification was introduced as part of the HITECH Act and is managed by the ASTP/ONC. The goal of certification is that a certified EHR has the capability to support meeting federal EHR usage regulations, like PI. So, the certification and PI regulations work together to define a set of functional requirements for an EHR. There have been multiple certification standards. The current one is the 2015 Edition.

Given the complexity of QPP/PI, APMs, and EHR functionality and the need for multi-stakeholder participation in the process, EHR certification is complex. Vendors of EHRs are the organizations who must understand certification in detail. It is helpful for providers to know how certification works at a high level so they can evaluate and converse with EHR vendors.

A summary of what the 2015 Edition covers is:



You can read about the details of the [Certification of Health IT](#) program at the HealthIT.gov website.

The best way to understand EHR certification is to look at an example. Certification requirements are tied to achieving particular EHR usage requirements under PI or APMs. They also define standards for how information in an EHR is represented (discussed in depth in a later lecture) and how testing for the criteria should occur. One example of this relationship is for the PI e-Prescribing requirement.

Example of PI e-Prescribing Requirement

PI Requirement

For at least one hospital discharge, medication orders for permissible prescriptions (for new and changed prescriptions) are queried for a drug formulary and transmitted electronically using Certified Electronic Health Record Technology (CEHRT).

Electronic Prescribing Certification Requirement

The related certification requirement works to ensure that this PI regulation can be met. It requires a certified EHR to meet regulations as described in [§ 170.205\(b\)\(3\) Electronic Prescribing](#).

Standards Criteria

Successful e-prescribing requires certain data representation standards. The required ones are defined at that same link.

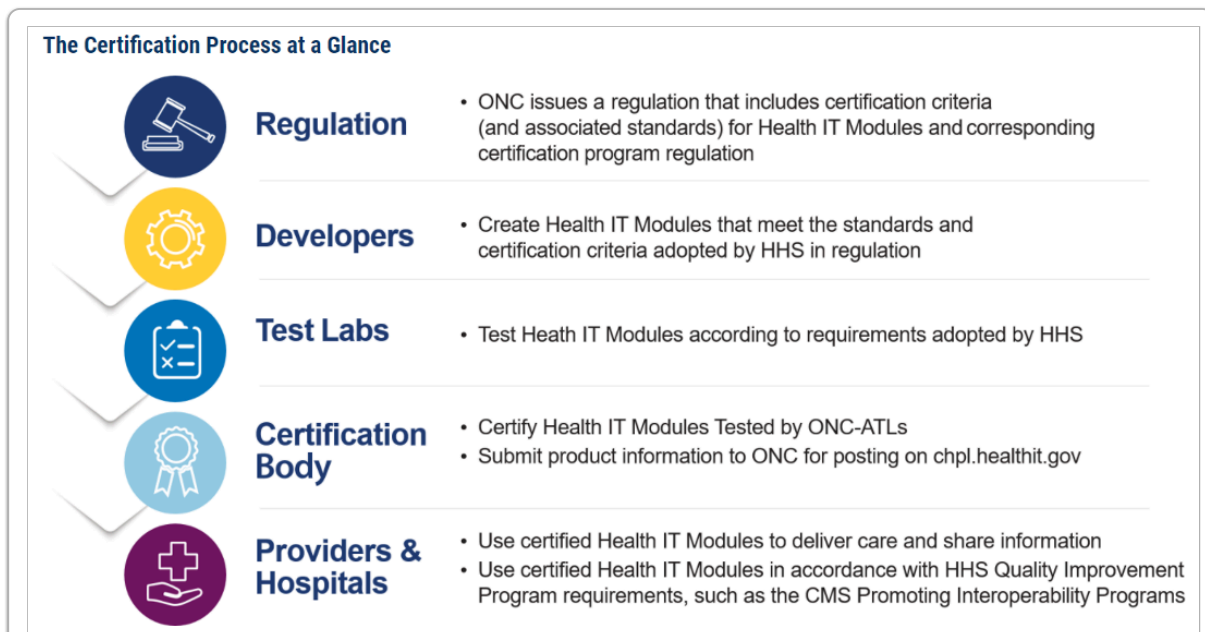
They will be discussed more in a later lecture.

Testing Procedure

The final component of certification is a defined testing procedure to see if an EHR meets the certification requirement. There is much detail to the testing procedure. The testing procedure for ePrescribe is defined at that same link.

The Cures Act added additional requirements that must be met to maintain certification. For more information see: [2015 Edition Cures Update – Base Electronic Health Record \(EHR\) Definition](#).

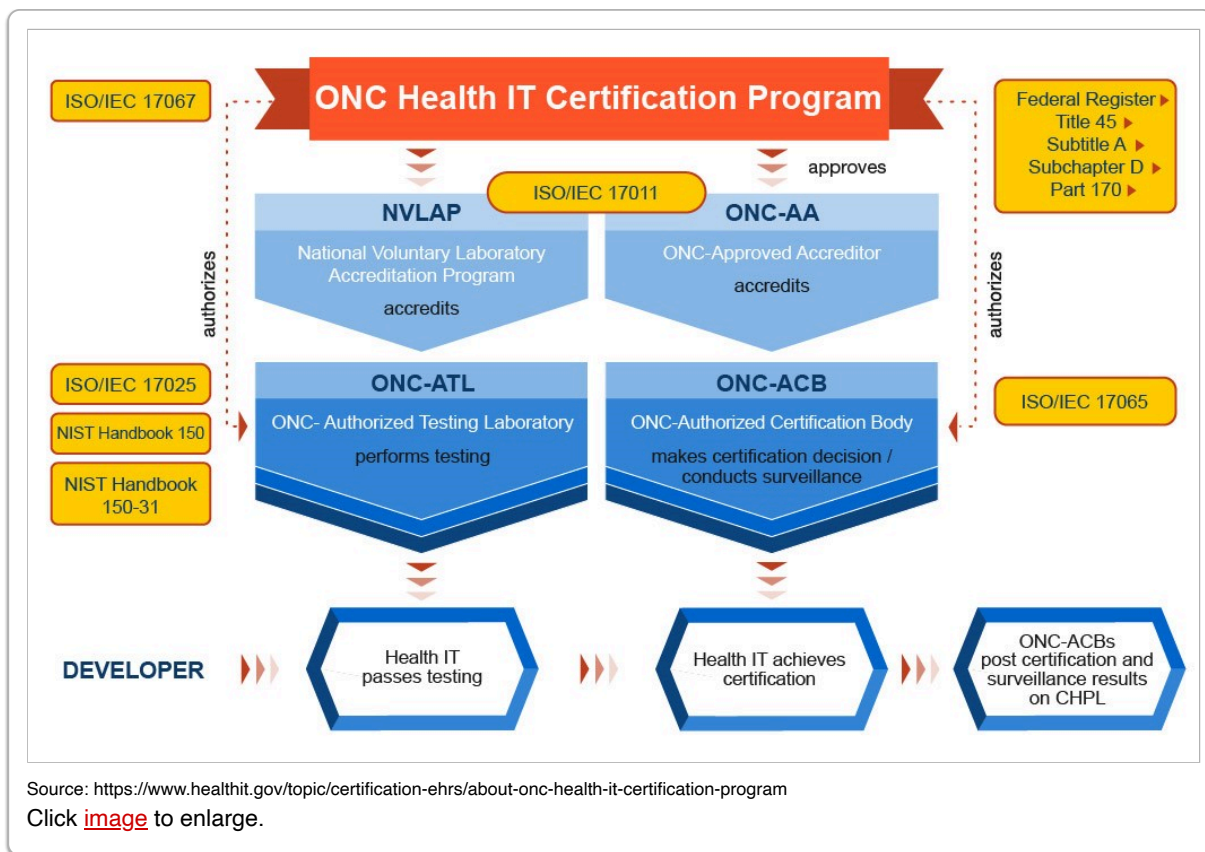
At a high level, the certification process is:



Source: <https://www.healthit.gov/topic/certification-ehrs/certification-process>

Click [image](#) to enlarge.

To make this program work requires not only the certification criteria but also a testing approach and infrastructure to implement the certification program. This infrastructure consists of multiple organizations as follows:



Currently Certified EHRs

Certified EHR systems are listed in the Certified Health IT Product List (CHPL) and are searchable multiple ways.

As part of the reading for this module, quickly review the [Certified Health IT Product List](#) from the HealthIT.gov website.

Summary

In summary, the federal government has made incenting use of EHRs an important policy goal. That effort started with the HITECH Act and Meaningful Use, which by both supporting and assisting in paying the cost of EHR deployment substantially increased the use of EHRs in the United States. The federal government continues to incent EHR use by tying it to how Medicare pays healthcare providers. This policy aims to reduce the growth in healthcare spending while maintaining and improving the quality of care. Use of EHRs and HIS will help achieve this goal. Part of the federal EHR programs is a certification program designed to ensure that EHRs can meet the regulations.

The regulations and certification goals are very complex and change often; students should keep up to date on the federal rules, including by watching the online resources referred to in this lecture.

Warning

- There are lots of details and fine print!
- The regulations change all the time!

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